PRODUCT REQUIREMENT DOCUMENT (PRD)

UMHackathon 2026

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Project Name: Z.AI Emergency Portal

Version: 1.0

1. Project Overview

- **Problem Statement**
  Flood prone regions, such as regions along Klang River and Klang Valley, face critical communication breakdowns during flooding disasters. Systems nowadays rely heavily on stable internet and unstructured information, which leads to delayed evacuations, panic, misinformation, inefficient rescue prioritization and difficult to locate victims when devices go offline. If connectivity and power fail, current solutions might stop working entirely and costing lives.

- **Target Domain**
  Our target domain is The Disaster Response and Emergency Management Systems, mainly focused on flood-prone communities & government and rescue agencies.

- **Proposed Solution Summary**
  Z.AI Emergency Portal is an AI-powered, offline-resilient disaster response system that is designed to function even when the infrastructure fails. This enables users to receive early evacuation notice and send structured SOS signals with minimal interaction. Users can stay connected through mesh networks even when the internet is down. At the same time, the rescue teams will be assisted with real-time prioritized rescue dashboards and AI-generated survival predictions and last-known-location tracking. This app does not rely on constant connectivity but is designed for the worse case scenario.

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2. Background & Business Objective

- **Background of the Problem**

  For years, disaster response systems relied on centralized infrastructure and stable communication networks. In a typical flood scenario, information flows in this way:

  1. Authorities issue warnings
  2. Citizen receive updates (if connected)
  3. Victims manually report their situation
  4. Rescue teams respond based on incomplete or delayed data.

  However, in flood-prone regions this model breaks down quickly, since the internet connectivity is often lost, power outages disable communication, victims panic and provide unclear or no information and the rescue teams do not have real-time visibility of the flood conditions. Hence, this system fails when it is needed the most.

- **Importance of Solving This Issue**

  It is crucial to solve this issue because disaster response is time sensitive and life critical. By improving the system, we can reduce the response time from hours to real-time, eliminate reliance on victims' input during panic situations, maintain communication independence of internet access and enable accurate rescue prioritization. This directly increases survival rates and reduces chaos during rescuing .

- **Strategic Fit / Impact**

  Z.AI Emergency Portal aligns with the global shift toward AI-driven decision systems, decentralized communication, and resilient infrastructure for climate disasters. This supports national and regional goals such as enhancing disaster prevention preparation in Malaysia, supporting agencies like *Jabatan Meteorologi Malaysia* through better data utilization, building smart, connected, and disaster-resilient communities. This product also aligns with SDG 11- Sustainable Cities and Communities to strengthen disaster resilience and protect vulnerable populations that are affected by flash floods in Malaysia.

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3. Product Purpose

3.1. Main Goal of the System

Our main goal is to create an AI-powered, disaster resilient response system that enables affected individuals to evacuate safely, provide crucial information and be located by the rescue team as soon as possible despite being offline and in extreme conditions.

3.2 Intended Users (Target Audience)

- **Primary Users**
    1. Residents living in flood-prone areas
    2. Families living in high risk zones\
- **Secondary Users**
    1. Disaster response teams
    2. Emergency rescue units
    3. NGOs and volunteers involved in relief efforts
- **Tertiary Users**
    1. Government agencies (disaster management authorities such as NADMA)
    2. Infrastructure and safety regulators

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4. System Functionalities

4.1. Description

Z.AI Emergency Portal operates as an AI-driven disaster response engine that transforms fragmented, real-time inputs into structured actionable rescue intelligence. Instead of relying on manual reporting or stable connectivity, the system continuously collects data from users, sensors and environmental signals, and AI will interpret, prioritize, and act on critical situations. It maintains functionality during offline conditions by using mesh communication.

4.2. Key Functionalities

- **Intelligent Alert & Evacuation Guidance**

  Triggers alerts when integrated systems such as *Jabatan Meteorologi Malaysia* issue natural disaster warning. (Simulated integration with Jabatan Meteorologi Malaysia for MVP.)

- **Real-time Route Adaptation**

  Users can mark blocked, destroyed, flooded paths with one tap, and the system will instantly update local device routes and shared network data when it is available. This ensures collective intelligence across users.

- **Structure SOS & Priority Engine**

  Users only require to give in simple inputs such as locations, water level and battery status, and AI will assign priority levels (P0-P3) based on urgency and generate standardized rescue tickets for the command centers.

- **Command Center Dashboard**

  All incoming rescue requests are displayed as structured task cards and automatically sorted by priority, time and status. This enables efficient rescue dispatch and tracking.

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4.3. AI Model & Prompt Design

4.3.1. Model Selection

Z AI GLM is used due to its ability to interpret unstructured human input under panic situations, convert incomplete data into structured emergency reports and perform real-time decision support under uncertainty. Traditional rule-based systems sometimes fail in an unpredictable disaster scenarios, while LLMs can provide flexibility, adaptability and contextual understanding

4.3.2. Prompting Strategy

This system adopts a multi-step prompting approach.

1. **Input Normalization:** Clean and interpret raw user input (button signals).
2. **Context Injection:** Combine user profile, sensor data and environmental conditions.
3. **Structured Output Generation:** Convert into standardized SOS format.

Inputs under disasters are often messy and incomplete, so a multi-step prompting can ensure accuracy and also consistency.

4.3.3. Context & Input Handling

The system supports short structured inputs which are button based and unstructured voice or text inputs by the victims. For large and complex inputs, the system uses chunking and prioritization, extracts only critical survival data such as location, risk level and battery level. If input exceeds limits, non necessary data will be discarded but core emergency signals are retained and processed.

4.3.4. Fallback & Failure Behavior

To ensure reliability, if the AI output is unclear or off-topic the system will default to predefined emergency templates. The retry mechanism will reprocess the inputs with simplified prompt structure. Besides, the rescue operators can also manually review and override the inputs. When the AI fails, the system will reduce to basic SOS transmission mode.

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5. User Stories & Use Cases

- **User Stories:**
  - **Example:** "As a victim stranded in a flood, I want to report my location and status using simple buttons so that rescue teams can find me instantly, even if I am panicking and cannot type."
- **Use Cases (Main Interactions):**
  - **SOS Submission Flow:** User inputs water depth and hazards via minimal UI buttons. The system appends GPS and battery data, then Z.AI GLM parses the intent and assigns a Priority Level (P0-P3) before sending it to the Rescue Dashboard.
  - **Offline Mesh Forwarding:** When the user loses internet connection, the app automatically switches to Store-and-Forward mode, saving the SOS packet in the Local Mesh Queue. The packet is synced automatically when a relay node is found.

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6. Features Included (Scope Definition)

- **Smart Alert & Evacuation System**
  Automatic disaster alerts will be triggered via simulated integration with Jabatan Meteorologi Malaysia for MVP . A personalized 15 minutes emergency preparation checklist is featured and the system provides offline escape maps & routes via downloadable PDF for zero-connectivity scenarios.

- **Offline Map & Navigation**
  For zero internet usage, the system features a pre downloaded map and guide users even during connectivity loss

- **Real time Hazard Reporting**
  The system features a one tap 'Route Blocked' feature to update routes for individual users and nearby users via sync when possible.

- **One tap SOS system**
  The system features a button-based emergency reporting which does not require typing and converts users input (location, water level, and situation) into structured rescue data.

- **AI-Powered Priority Engine**
  The system automatically assigns rescue priority level (P0-P3) and processes sensor data and user input, ensuring the most critical cases are rescued first.

- **Mesh Network Communication**
  Device to device communication via Bluetooth or WiFi enables SOS data transmission without mobile network. (Note: For the Hackathon MVP, the physical hardware mesh is simulated via a Local Store-and-Forward Queue mechanism within the app interface).

- **"The Dead Phone" Protocol**
  When the device is severely destroyed by the flooding situation or low battery, the last known location will be automatically uploaded before disconnection and an emergency data packet will be sent when the battery level is <5%, low power mode will be activated and there will be LED beacon for physical rescue visibility.

- **Rescue Command Dashboard**
  The system displays all victims as structured task cards, which are auto-sorted by priority, time and status. This enable efficient dispatch and tracking of the victims.

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- **Minimalist Emergency UI**
  The app will be in dark mode, with a low-distraction interface and fully button-driven interaction. There will be no unnecessary animations in order to conserve battery.

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7. Features Not Included (Scope Control)

- **Full-scale Infrastructure Deployment**
  There is no real time integration with national disaster infrastructure and sensor networks are assumed already exist, not built within this project.

- **Guaranteed Long-Range Communication**
  Mesh Network is limited to short range device to device transmission. Hence, this does not guarantee rapid delivery over long distances without relay nodes.

- **Fully Autonomous Rescue Dispatch**
  AI only supports the part of decision making and prioritization, the final rescue actions are still controlled by human operators.

- **Other Disasters support**
  This system is currently optimized for flood scenarios only, it does not yet support other disasters such as earthquakes and wildfires.

- **Large-Scale Concurrent Load Optimization**
  High volume real time synchronization across millions of users is not fully optimized in this version. The current version is only designed for regional-scale deployment in early stages.

- **Advanced Data Governance & Compliance**
  Full compliance with large scale data regulations is not fully implemented. Basic user data handling is included, but enterprise grade security layers will be implemented in the future.

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8. Assumptions & Constraints

We assume users have pre-installed the app and enable essential permissions (location services and bluetooth) before flooding disaster occurs.

- **LLM Cost Constraint**
  Estimated cost per user session is low to moderate, since the interactions are short and structured, most AI processing focuses only on SOS structuring and priority classification. Instead of free-text input, the system uses button driven UI and applies event-triggered AI activation only when it is needed. This significantly reduced token usage compared to continuous AI interaction systems.

- **Technical Constraints**
  Depends on smartphone hardware capabilities such as GPS, Bluetooth/WiFi Direct and Barometer which is not available on all devices. The mesh network is limited by device compatibility and OS restrictions. Moreover, integration with agencies such as Jabatan Meteorologi Malaysia assumes API availability and reliability.

- **Performance Constraints**
  The AI processing is event based and not continuous, hence the real time responsiveness may be affected by device performance, battery level and network availability. Mesh communication may also introduce latency due to multi-hop transmission. This system prioritizes reliability over speed in extreme conditions.

- **User Input Constraints**
  The system is designed for minimal interactions without complex input, so it relies on predefined buttons and simple selections. But there might be chances where users misreport or press wrong inputs under panic situations and cannot provide clear voice input in a noisy and chaotic environment. AI simplifies and validates users' inputs.

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9. Risks & Questions Throughout Development

- **AI Misinterpretation Risk**
  Users may press the wrong button or give unclear voice input during panic situations, and AI may incorrectly classify urgency or situation. As a mitigation, the system will feature structured button inputs as primary interaction, allowing edit or confirmation of SOS before submission .

- **User Adoption Risk**
  There is a risk where the users did not install or properly set up the app when the disaster already occurred. As a mitigation there should be awareness campaigns, integration with government initiatives, universities or communities.

- **Battery Drain / Device failure**
  There is a risk where the users' device is in low battery level or severely destroyed during the disaster event. As a result users may lose device power before the arrival of the rescue team. To mitigate, low power mode will be activated which disables non-essential functions and an emergency packet (last known location and status) will be sent when the battery level is <5%.

- **Data Synchronization Issues**
  Users might send conflicting updates such as one says a particular road is blocked while another one says the road is still usable. The system will use timestamp-based validation and accept majority consensus from nearby devices, so that it can mark uncertain data as unverified.

- **Mesh Network Reliability**
  When there are not enough nearby devices to relay signals, SOS messages may be delayed or not delivered, hence breaking the communication chain. As a mitigation the system implements a Store-and-Forward Mechanism, where the device will store the message and send it to another device when a connection becomes available and prioritize critical data packets only.

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